Dr Hengfei Wang

POSTDOCTORAL RESEARCH FELLOW

Perception and Activity Understanding Group, EPFL & Idiap, Switzerland

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About Me

I am a Postdoctoral Research Fellow at EPFL and Idiap Research Institute, working with Jean-Marc Odobez on Human Perception and Activity Understanding. I received my PhD in Computer Science from the University of Birmingham in 2025. Prior to that, I earned both my Bachelor's and Master's degrees in Mechanical Engineering from Tsinghua University in 2017 and 2020, respectively.

My research spans various areas of artificial intelligence, including computer vision, multimedia, and human-computer interaction. My long-term research interests lie in understanding and generating human behavior. Specifically, I leverage deep learning techniques to track human motion (gaze, hand, and body), analyze human behaviors, and synthesize realistic digital humans.

Education

University of Birmingham (UOB) — *Birmingham, UK*

PH.D. IN COMPUTER SCIENCE — *Sep. 2020 - Mar. 2025*

Tsinghua University (THU) — *Beijing, China*

B.ENG. & M.ENG. IN MECHANICAL ENGINEERING — *Sep. 2013 - Jun. 2020*

Employment

Postdoctoral Research Fellow — *EPFL & Idiap*

PERCEPTION AND ACTIVITY UNDERSTANDING GROUP — *Sep. 2025 - Present*

- **Human Perception and Activity Understanding**
- **Keywords**: Gaze Estimation/Following
- **Introduction:** Coming soon...
- **Line Manager:** Dr. Jean-Marc Odobez

Postdoctoral Research Fellow — *University of Birmingham*

SCHOOL OF COMPUTER SCIENCE — *Mar. 2025 - Sep. 2025*

- **Gaze Estimation in the Wild**
- **Keywords**: Gaze Estimation
- **Introduction:** This research focuses on developing robust gaze estimation models that perform reliably in real-world, unconstrained environments. Unlike controlled lab settings, real-world scenarios involve varying head poses, lighting conditions, occlusions, and diverse demographics, which pose significant challenges for accurate gaze tracking. The project aims to bridge this gap using deep learning, domain adaptation, and data augmentation techniques. I am responsible for designing and implementing novel gaze estimation algorithms, conducting large-scale experiments, and contributing to publications. I also collaborate with other researchers to integrate the models into broader human-computer interaction systems.
- **Line Manager:** Prof. Hyung Jin Chang

Professional Activities

ACADEMIC EVENT

2025	**Program Committee**, 1st Human-Autonomous Vehicle Interaction Workshop	*WACV Workshop*
2024	**Webchair**, 6th International Workshop on Gaze Estimation and Prediction in the Wild	*CVPR Workshop*
2023	**Webchair**, 5th International Workshop on Gaze Estimation and Prediction in the Wild	*CVPR Workshop*
2022	**Webchair**, 4th International Workshop on Gaze Estimation and Prediction in the Wild	*CVPR Workshop*
2021-2025	**Maintainer of Biggest Gaze Research Repository**, Awesome Gaze Estimation in Github	*Github Repository*

TEACHING

2025	**Assistant Supervisor**, Youngchan Choi (PhD Student): 3D-Aware Gaze Redirection	*University of Birmingham*
2019	**Teaching Assistant**, Fundamentals of Testing and Inspection Technology (Undergraduate Course)	*Tsinghua University*

GRANT

2018-2020	**Participants [£411,552]**, "In-line Inspection of Urban Gas Pipeline Networks"	*Beijing Municipal Science & Technology Commission*

Peer-Review Activities

Workshop

International Workshop on Gaze Estimation and Prediction in the Wild (2023/2024)
Human-Autonomous Vehicle Interaction Workshop (2025)

Conference

CVPR(2021/2022/2023), ECCV(2022/2024), AAAI(2024/2025), ICME2025, NeurIPS2025, ACM MM2025

Journal

IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

Honors & Awards

2024	**ECCV Oral Paper Award**, ECCV2024	*MiCo Milano, Italy*
2020-2024	**China Scholarship Council (CSC) Scholarship**, China Scholarship Council [£57,600]	*Beijing, China*
2023	**BMVC Student Funding**, BMVC2023	*Aberdeen, UK*
2023	**BMVC Oral Paper Award**, BMVC2023	*Aberdeen, UK*
2019	**ICRAE Best Oral Presentation Award**, ICRAE2019	*Singapore*
2018	**Outstanding Social Work Scholarship**, Tsinghua University	*Beijing, China*
2017	**Finalist of Comprehensive Winter Study Abroad Program**, University of Tokyo	*Tokyo, Japan*
2016	**Finalist of International Undergraduate Research Experience Program**, University of Notre Dame	*Notre Dame, USA*
2014	**China National Encouragement Scholarship**, Chinese Government	*Beijing, China*
2014	**Science and Technology Innovation Scholarship**, Tsinghua University	*Beijing, China*
2013	**2nd Prize in Mechanical Innovation Competition**, Tsinghua University	*Beijing, China*

Presentations

2024	**Generalization of Gaze Estimation**, Invited Talk @ 2024 CVPR Digestion Workshop in UOB	*Birmingham, UK*
2023	**High-Fidelity Eye Animatable Neural Radiance Fields for Human Face**, BMVC2023	*Aberdeen, UK*
2023	**Image Synthesis for Gaze Estimation**, Invited Talk @ 2023 CVPR Digestion Workshop in UOB	*Birmingham, UK*

Publications

International Conference

2025
Roll Your Eyes: Gaze Redirection via Explicit 3D Eyeball Rotation
YoungChan Choi*, Hengfei Wang*, Yihua Cheng, Boeun Kim, Hyung Jin Chang, Younggeun Choi, Sang-Il Choi
ACM International Conference on Multimedia (ACM MM)

2025
3D Prior Is All You Need: Cross-Task Few-shot 2D Gaze Estimation
Yihua Cheng, Hengfei Wang, Zhongqun Zhang, Yang Yue, Boeun Kim, Feng Lu, Hyung Jin Chang
Conference on Computer Vision and Pattern Recognition (CVPR)

2024
TextGaze: Gaze-Controllable Face Generation with Natural Language
Hengfei Wang, Zhongqun Zhang, Yihua Cheng, Hyung Jin Chang
ACM International Conference on Multimedia (ACM MM)

2024
NL2Contact: Natural Language Guided 3D Hand-Object Contact Modeling with Diffusion Model
Zhongqun Zhang, Hengfei Wang, Ziwei Yu, Yihua Cheng, Angela Yao, Hyung Jin Chang
European Conference on Computer Vision (ECCV), Oral Presentation

2023
High-Fidelity Eye Animatable Neural Radiance Fields for Human Face
Hengfei Wang, Zhongqun Zhang, Yihua Cheng, Hyung Jin Chang
The British Machine Vision Conference (BMVC), Oral Presentation

2023
GazeCaps: Gaze Estimation With Self-Attention-Routed Capsules
Hengfei Wang, Jun O. Oh, Hyung Jin Chang, Jin Hee Na, Minwoo Tae, Zhongqun Zhang, Sang-Il Choi
Conference on Computer Vision and Pattern Recognition Workshop (CVPRW)

2019
Robotic System with Power Line Communication for In-pipe Inspection of Underground Urban Gas Pipeline
Hengfei Wang, Zandong Han, Qianxian Ma
International Conference on Robotics and Automation Engineering (ICRAE), Best Oral Presentation